Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2024 | ENGI3331-WE01/<br>ENGI3481-WE01 |

**Title: (BEng/MEng)**

Semiconductor Physics and Devices/Power Semiconductor Devices

| | |
|---|---|
| Time Allowed: | 2 hours |
| Additional Material provided: | Yes - Formula Sheet included as part of exam paper. |
| Reporting problems during the exam: | Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk.<br><br>If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk. |

| | |
|---|---|
| **Instructions to Candidates:** | This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable.<br><br>Answer ALL questions.<br><br>Please answer each question on a new page.<br><br>All workings must be shown.<br><br>This is an online open-book exam.<br><br>• *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.*<br><br>• *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.* |

| Revision: | |
|---|---|

**Question 1**

You are investigating the response of a slab of gallium arsenide (GaAs) ($E_g$ = 1.42 eV, $n_i = 1.8 \times 10^6$ cm$^{-3}$) to optical generation of carriers. The slab, shown schematically in **Fig. Q1**, has a length of $L$, and is sufficiently thin to allow uniform light generation throughout its bulk. It is uniformly doped with donor density of $N_D = 8 \times 10^{17}$ cm$^{-3}$. Our intention is to generate an excess of carriers that constitutes 'low injection' conditions. The solutions to two useful differential equations are given at the bottom of this page.

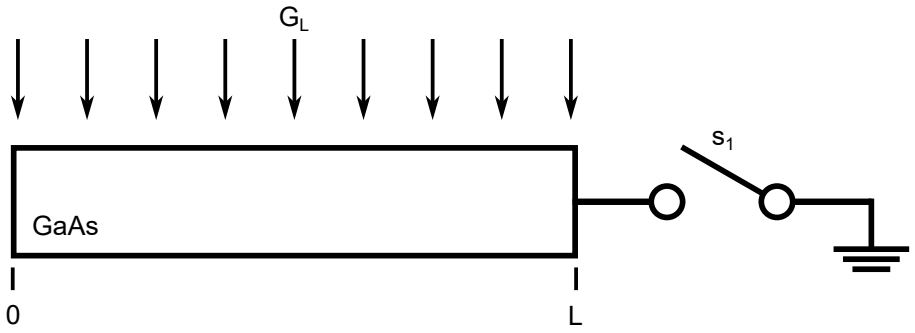



**Figure Q1**

a. Explain what is the meaning of 'low injection'. How is it expressed in the continuity equation?

**[12%]**

b. What is the range of wavelengths that can be absorbed in GaAs? Give your answer in nanometres.

**[16%]**

At time $t = 0$, the illumination is turned on producing a uniform generation rate within the entire volume of the semiconductor of $G_L = 1.25 \times 10^{22}$ cm$^{-3}$s$^{-1}$. The density of excess carriers at $t$ = 15 ns is 85% of its steady-state density.

c. i. Derive an expression for the density of excess carriers, $\Delta p$, as a function of time.

ii. Use the expression you have just found in **Sec. c. i.** to calculate the density of excess carriers in steady-state conditions. Give your answer in cm$^{-3}$.

**[40%]**

We now ground one side of the slab by closing switch $s_1$, forcing $\Delta p(L) = 0$. Under these conditions, the density of excess carriers several diffusion lengths away from the ground remains unchanged.

d. i. Calculate $L_p$ if the hole mobility in GaAs is $\mu_p$ = 400 cm$^2$V$^{-1}$s$^{-1}$. What would be the density of excess carriers at $x = 0$ if the length of the slab is 30 $\mu$m? If you did not solve **Sec. c. ii.** you may use $\tau_p = 10^{-8}$ s.

ii. Derive the steady state spatial profile of excess carriers, $\Delta p(x)$, and show that it is proportional to

$$1 - \frac{\sinh(x/L_n)}{\sinh(L/L_n)}$$

Clearly state the assumptions you made in your derivation.

**[32%]**

You may find these differential equations and their solutions useful in your working:

| **Diff. Equation** | **Solution** |
|---|---|
| $y' = A - \frac{y}{B}$ | $y = A \cdot B + K_1 e^{-x/B}$ |
| $y'' = \frac{y}{C} - D$ | $y = K_2 e^{x/\sqrt{C}} + K_3 e^{-x/\sqrt{C}} + C \cdot D$ |

**Question 2**

An error in quality assurance led a semiconductor fabrication plant to produce a highly asymmetric $pn^+$ junction diode instead of the intended lightly-doped symmetric diode. To characterise this unknown device, a reference diode is connected to it in series, in the manner described in **Fig. Q2**. The reference diode has a saturation current $I_s^{(0)} = 3 \times 10^{-14}$ A. Both diodes are made of silicon ($E_g = 1.12$ eV, $n_i = 10^{10}$ cm$^{-3}$, $\varepsilon_r = 11.7$), and have a cross-section of $10^{-4}$ cm$^2$

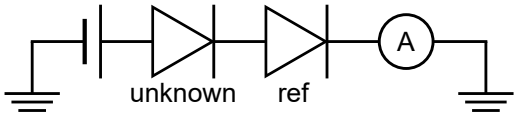



**Figure Q2**

a. A current of 13 μA is measured under an applied bias of 1 V across the entire circuit. Assume that there are no other resistive elements in the circuit and calculate the unknown device saturation current, $I_s^{(u)}$.

**[36%]**

Next, the reference diode is disconnected and the capacitance of the unknown diode is measured at zero applied bias and at a reverse bias of 0.5 V. The zero-bias capacitance is found to be $C_0 = 1.40$ pF and the reverse bias capacitance is $C_r = 1.11$ pF.

b. i. Calculate the built-in potential, $V_{bi}$ of the unknown diode. Give your answer in volts.

ii. Calculate the doping profile on both sides of the junction. Give your answer in cm$^{-3}$.

**[36%]**

c. i. Calculate the electron diffusion length, if its mobility is $\mu_n = 1,400$ cm$^2$V$^{-1}$s$^{-1}$. Give your answer in micrometres. If you did not solve the previous sections, you may use $I_s^{(u)} = 1.8 \times 10^{-13}$ A, $N_A = 9 \times 10^{14}$ cm$^{-3}$, and $N_D = 10^{19}$ cm$^{-3}$.

ii. Calculate the electron lifetime. Give your answer in nanoseconds.

**[28%]**

**Question 3**

You have been asked by the Chief Technology Officer of a local Semiconductor Company to design their next generation of semiconductor devices to support the development of fusion energy in the UK. The challenge is to design a p-i-n diode and enhancement mode JFET that can be manufactured from the same silicon carbide wafer.

a. Determine the dopant concentration of the drift region to give a maximum electric field of 865 kV/cm.

**[20%]**

b. Under the assumption that the diode is a non-punch-through structure, determine the thickness of the drift region required to enable the p-i-n diode to block 200 V. You should assume that the thickness of the p+ region is 0.25 $\mu$m.

**[20%]**

c. Determine the on-state resistance of the diode and hence determine the total forward voltage drop and power dissipation for a device rated for 250 A·cm$^{-2}$.

**[28%]**

d. The diode is rated to conduct 10 A at the forward voltage calculated in question 3.c. Having determined the area of the diode, calculate the charge injected in to the diode when it transitions from blocking to zero bias.

**[32%]**

**Question 4**

The structure of the wafer to be used to create the vertical power JFET designed in **Question 3** is show in **Fig. Q4**. If you did not successfully determine the dopant concentration and thickness of the drift region in **Question 3**, use the assumption that $N_D = 3 \times 10^{16}$ cm$^{-3}$ and $t_{epi} = 4$ μm.

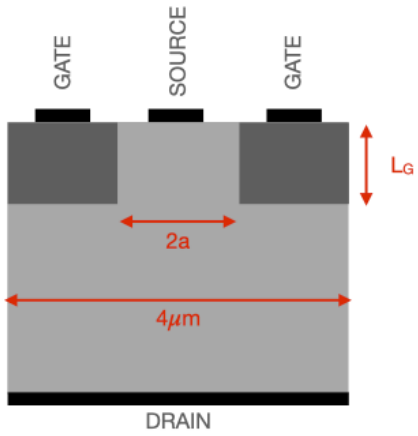



**Figure Q4**

a. Determine the depth of the implantation used to form the gate region, which is denoted by $L_G$ in **Fig. Q4**, so that the gate drain breakdown voltage is 100 V.

**[20%]**

b. The JFET is to operate in enhancement mode with a threshold voltage of 1.5 V. Calculate the channel width, denoted as 2*a* in **Fig. Q4**, to enable this.

**[24%]**

c. Calculate the on-state resistance of the JFET for the device dimensions shown in **Fig. Q4** and hence the current when $V_{DS} = 1$ V. The gate - source voltage is 2.5 V.

**[24%]**

d. The completed JFET comprises a number of devices in parallel to enable operation at 10 A and 400 °C. The gate width of the JFET is 1 mm. Calculate the number of parallel devices required to achieve this and the resistance at 300 K.

**[32%]**

**END**